SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting

Pranav Asthana, Alex Hanson, Allen Tu, Tom Goldstein, Matthias Zwicker, Amitabh Varshney

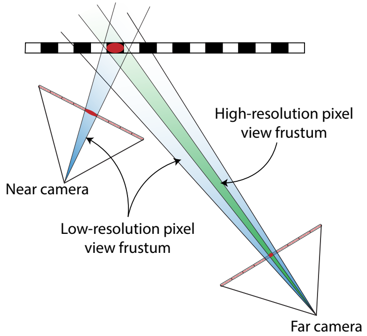
splatsure.github.io

Contact: pasthana@umd.edu

CVPR DENVER COLORADO JUNE 3-7, 2026

Motivation

Can we use 2D super-resolution to build high-resolution 3DGS models but also remain consistent to the ground truth?

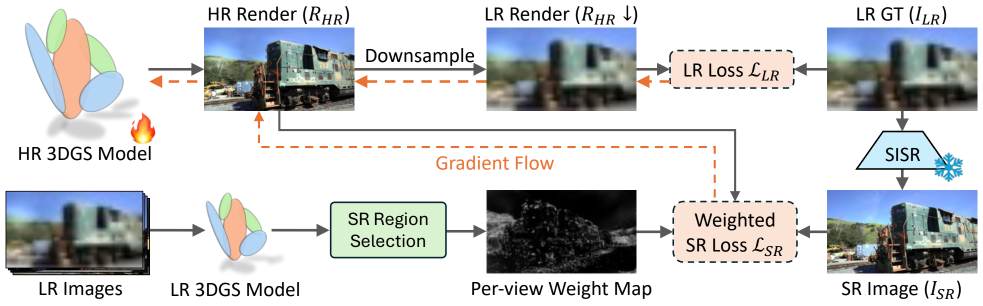


Images of a scene do not sample 3D content uniformly

Near cameras can provide high-resolution information for rendering distant views

We can use this disparity in captured frequency across views to inform where super-resolution is really required

Method



Gaussian Fidelity Score

ρ = Ratio of Gaussian's max to min sceen-space radius across views

$\rho \approx$ Disparity in sampling across views

High $\rho \rightarrow$ High variance in sampled frequency

Low $\rho \rightarrow$ Uniform sampling

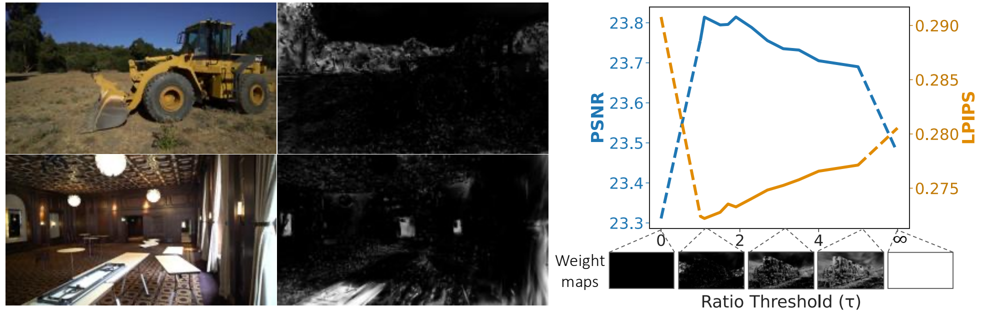
$$\text{score} = \text{Sigmoid}\left(\frac{\rho-\tau}{k}\right)$$ where τ is a threshold and k controls smoothness

Super-Resolution Region Selection

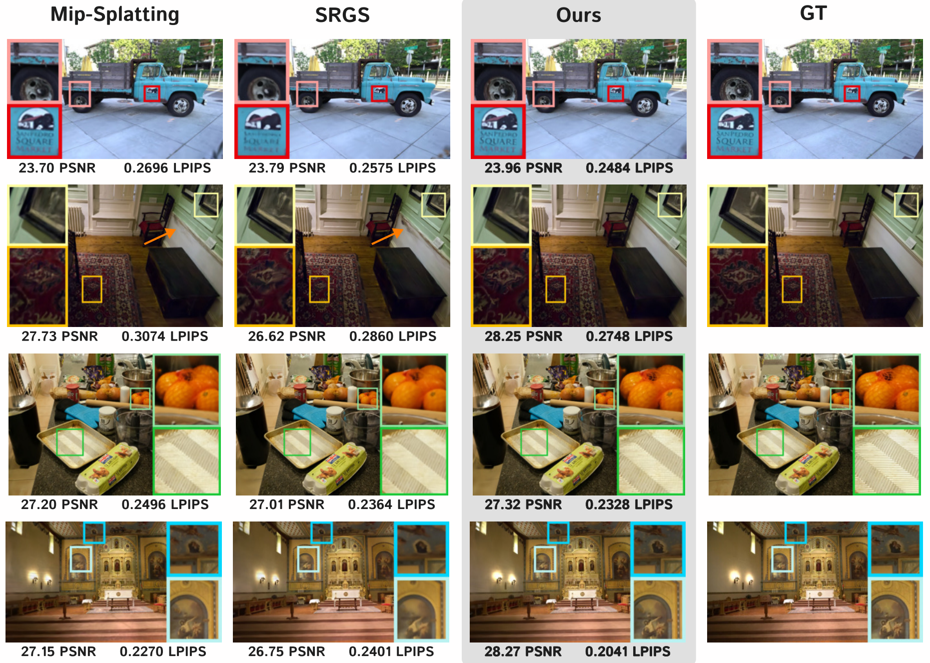
Convert **per-Gaussian** score to **per-view** weight map for super-resolution selection

$$W_t = \underbrace{\left(1 - \text{Render}(\text{score}_{\mathcal{G}})\right)}_{\text{Regions with low fidelity scores}} + \underbrace{\text{Render}\left(\mathbf{1}_{\mathbf{M(t)}}(\mathcal{G})\right)}_{\text{Regions closest to current view}}$$

$\mathcal{L}_{\text{SR}}$ is spatially weighted by this per-view weight map W_t



Results



We achieve consistently sharper novel renders and improved consistency with the ground truth by selectively applying super-resolution!

Method	Tanks & Temples (rendered at 4× input resolution)							
	SSIM ↑	PSNR ↑	LPIPS ↓	FID ↓	CMMD ↓	DreamSim ↓	MUSIQ ↑	NIQE ↓
3DGS (LR)	0.669	19.41	0.350	71.58	2.013	0.0895	57.776	3.412
3DGS + StableSR	0.751	22.47	0.300	59.29	1.123	0.0667	56.748	4.945
Mip-Splatting	0.767	23.10	0.303	52.46	1.137	0.0597	46.571	5.043
SRGS + StableSR	0.771	23.32	0.286	49.11	1.048	0.0535	55.209	4.633
Ours + StableSR	0.784	23.81	0.272	37.72	1.040	0.0413	58.332	3.928

* Please see our website and paper for additional qualitative and quantitative results

Acknowledgements

This work was made possible by NSF Grants 21-37229 and 22-35050, DARPA TIAMAT, and the NSF TRAILS Institute (2229885). This research is based upon work supported by the Office of the Director of National Intelligence (ODNI), Intelligence Advanced Research Projects Activity (IARPA), via IARPA R\&D Contract No. 140D0423C0076. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the ODNI, IARPA, or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright annotation thereon. Additional support was provided by Coefficient Giving.